

6. (a) **S** and **T** are two metal halide compounds. The ions in each compound were identified by observations from a flame test and the addition of silver nitrate solution.

(i) Complete the table.

[4]

Compound	Flame test colour	Symbol of ion	Observation on adding silver nitrate solution	Symbol of ion
S	brick red		yellow precipitate	
T		Ba ²⁺	white precipitate	

- (ii) Suggest why a silver nitrate test would not be able to distinguish clearly which halide ions are present in a **mixture** of compounds **S** and **T**.

[1]

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- (b) Write an **ionic** equation for the formation of the precipitate in the reaction between sodium chloride solution and silver nitrate solution.

Include state symbols.

[3]

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